A Network Analysis of the Cryptocurrency Market

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***Abstract*—In this study, we conduct a network analysis with centrality measures, using historical daily close prices of top 120 cryptocurrencies between 2013 and 2020, to study and understand the dynamic evolution and characteristics of the cryptocurrency market. Our study has two primary findings: (1) the overall cross-return correlation among the cryptocurrencies is weakening from 2013 to 2016 and then strengthening thereafter; (2) cryptocurrencies that are primarily used for transaction payment, notably BTC, dominate the market until mid-2016, followed by those developed for applications using blockchain as the underlying technology, particularly data storage and recording such as MAID and FCT, between mid-2016 and mid-2017. Since then, ETH, alongside with its strongly correlated cryptocurrencies have replaced BTC to become the benchmark cryptocurrencies. Furthermore, during COVID-19, QTUM and BNB have intermittently replaced ETH to take the leading positions due to their active community engagement during the pandemic.**

Keywords—Cryptocurrency market, network analysis, Centrality measures

I. Introduction

Over the past two decades, there has been considerable work on how network analysis can be used to better understand the financial markets [1]. Network can be used to model the interactions among different financial assets such as stocks, currencies and commodities. One of the prominent techniques is the use of correlation-based network to understand and forecast the dynamics in the financial markets. In a network, nodes and edges represent financial assets and their correlations, respectively. Then, network construction methods are used to construct less complex networks that typically include important nodes and edges, e.g. those with strong correlation. With these networks, centrality measures can be used to identify important or influential nodes or assets that play leadership roles in the financial markets.

The aim of this paper is to apply network analysis with centrality measures to analyze the cryptocurrency market, which has been receiving increased attention in recent years. Using historical daily close price of top 120 cryptocurrencies between 2013 and 2020, we aim to understand the current cryptocurrency market by studying its: (1) correlation structure, dynamic evolution and characteristics; (2) stability of the cryptocurrency network that models the market; (3) most influential cryptocurrencies during different stages of the evolution. In addition, based on our empirical results, we draw some conclusions for investors who are interested in a short-term trading strategy in the cryptocurrency market.

II. Literature Review

Network analysis can be used to identify important or influential nodes in a network. It has been extensively employed to identify influential stocks, currencies, and commodities in many different corresponding markets across the world [1, 19]. With respect to the cryptocurrency market, the work in [2] conducted a correlation analysis and discovered that those that were similar in their codebases were positively correlated. The work in [3] used network analysis to analyze 16 cryptocurrencies using the Minimum Spanning Tree (MST) and hierarchical tree methods. The evolution of the cryptocurrency market is also studied in [5]. However, all their analysis focused either on a specific (or spliced) time period or on a specific set of cryptocurrencies, which did not investigate the time-varying or dynamic evolution of the influential cryptocurrencies over a longer period of time as the cryptocurrencies are entering the market at different times. In this study, we adopt a network construction method together with five centrality measures to investigate the dynamic evolution and characteristics of the cryptocurrency market over seven years between 2013 and 2020. We further study which cryptocurrencies are influential during which periods of the evolution, including the COVID-19 pandemic, as well as draw some conclusions for investors who are interested in a short-term trading strategy in the cryptocurrency market.

III. Dataset

The dataset used in this study is the time series of daily close prices of top 120 cryptocurrencies by market capitalization as of 14.08.2020. This sample of cryptocurrencies can be considered as a good representative of the cryptocurrency market since it represents more than 90% of the total cryptocurrency market capitalization as of this date. Our data covers the period from 29.04.2013 (i.e. the first day when BTC close price becomes available in the dataset) to 14.08.2020, collected from http://coinmarketcap.com [17]. There are up to 2665 daily returns for each cryptocurrency time series data during the period investigated as the cryptocurrencies entered the market at different times. All the cryptocurrency prices are quoted in USD. As time series data may contain trends and seasonality, they may need to be removed prior to modeling. In addition to this, since the time series data of many cryptocurrencies is non-stationary, we therefore transform the prices into logarithmic returns using the formula:

$$r_i(t) = ln\, p_i(t) - ln\, p_i(t - \Delta t) \tag{1}$$

where $r_i(t)$ and $p_i(t)$ are the return and the close price of cryptocurrency i at day t, respectively. In our case, $\Delta t = 1$ day. Another reason for this transformation is to make $r_i(t)$ independent of the price scale, e.g. a cryptocurrency change from 10 to 11 BTC will have the same logarithmic difference as a change from, 500 to 550 BTC. This makes a comparison among cryptocurrencies at different price scales possible.

IV. Methodology

A. Cross-return Correlation Coefficient Matrix

The input to construct a network for analysis is the return correlation among all pairs of cryptocurrency. We first create a cross-return correlation coefficient matrix using the Pearson

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product-moment correlation for every pair of the cryptocurrency i and j as follows:

$$c_{ij} = \frac{\langle r_i . r_j \rangle - \langle r_i \rangle \langle r_j \rangle}{\sqrt{\left(\langle r_i^2 \rangle - \langle r_i \rangle^2\right)\left(\langle r_j^2 \rangle - \langle r_j \rangle^2\right)}} \quad (2)$$

where r_i represents the return vector of cryptocurrency i of a time period (i.e. 182 days or 6 months in this study as we will evaluate the impact of COVID-19 on the cryptocurrency market with the length of each time window being the current length or duration of the pandemic since its outbreak. The value of each cross-return correlation coefficient matrix item ranges from -1 to 1. When the value approaches 1, the two corresponding cryptocurrencies move similarly, i.e. when the return of one increases, then the return of the second increases similarly. On the contrary, when the value approaches -1, the movement of the two cryptocurrencies vary inversely (i.e. increases the first means decreases the second). When the value is 0, prediction may not be possible as the two cryptocurrencies vary in a random way. The result of the calculation is a N x N cross-return correlation coefficient matrix where N is the total number of the cryptocurrencies (i.e. N = 120 in this study), and the diagonal of the matrix is equal to 1 as any cryptocurrency has a perfect correlation with itself.

B. Network Construction

When constructing a network, a node represents a cryptocurrency while an edge represents the return correlation between the connected cryptocurrency node pair. A complete or fully connected network is then constructed, which often contains a large amount of redundant information. Several network construction methods, such as Minimum Spanning Tree (MST), Threshold Network (TN) and Plane Maximum Filter Graph (PMFG), have been proposed to reduce the network complexity by including meaningful nodes and edges only for studying correlations of financial time series (e.g. edges with strong correlation). However, the aim of our work is not to investigate and compare which method performs better than the others but to exploit their own advantages to filter pertinent information from a complex network modeling the return correlation for all pairs of the cryptocurrencies (a detailed review of these network construction methods is discussed in [6]). For the rest of this paper, we adopt MST to construct cryptocurrency networks as it is simple and includes all the cryptocurrencies for our analysis. Nevertheless, we have also implemented and performed our evaluation using the networks constructed by TN and PMFG, and reached similar results and findings to the MST, thus making our conclusions more reliable.

With the cross-return correlation coefficient matrix, we construct an undirected network using MST. Studying the pattern of connections in the networks would provide helpful information about the internal structure of the cryptocurrency market. Constructing the MST network requires a variable to measure the distance, satisfying the three axioms of Euclidean distance:

- **Positive Definiteness**: $d_{ij} = 0 \; if \; and \; only \; if \; i = j$
- **Symmetry**: $d_{ij} = d_{ji}$
- **Triangular Inequality**: $d_{ij} \leq d_{ik} + d_{kj}$

The distance that satisfies that above three axioms is derived by the work in [7]:

$$d_{ij} = \sqrt{2\left(1 - c_{ij}\right)} \quad (3)$$

where d_{ij} denotes the distance between nodes i and j. The distance function measures a distance between a pair of cryptocurrencies using their return correlation coefficient. The mapping between the return correlation coefficient and the distance function ranges from 0, for totally correlated cryptocurrencies (i.e. when the return correlation coefficient is equal to 1), to 2 for totally anti-correlated cryptocurrencies (i.e. when the return correlation coefficient is equal to -1). For uncorrelated cryptocurrencies, the distance is $\sqrt{2}$. This transformation creates a N x N distance matrix from the N x N cross-return correlation coefficient matrix. We use the method outline in [7] for constructing the MST network, which is a graph constructed by linking N nodes with $N - 1$ edges without loops such that the sum of all edge distance is the minimum, i.e., the MST network uses the $N - 1$ linkage to extract the most important information from the cross-return correlation coefficient matrix.

C. Moving Time Window Analysis

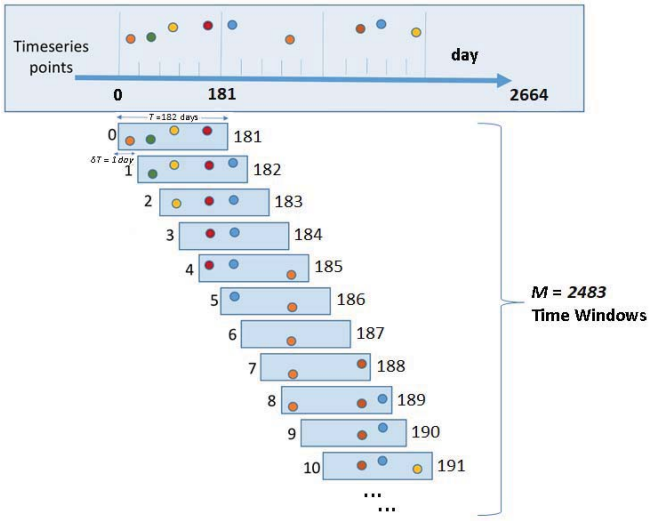

Fig. 1. Moving time window analysis

We adopt moving time window approach, as shown in Figure 1, to study the evolution of the cryptocurrency network over a period of 7 years. The reason for using this approach is that cryptocurrencies are progressively entering the market at different times and we avoid restricting our work to only a specific set of cryptocurrencies in a specific time period when their close prices become completely available. The question to be researched is which cryptocurrencies are influential during which periods of time. To look at the affiliation between the return of cryptocurrency i and j, we calculate the cross-return correlation coefficient matrix for all trading days in the entire studied time period. The data is divided timewise into M windows t = 1, 2, …, M of width T corresponding to the number of daily return included in each window. Several consecutive windows overlap with each other, the extent of which is dictated by the window step length parameter δT, describing the displacement of the window, measured also in trading days. The choice of window width is a trade-off between too noisy and too smoothed data for small and large window widths, respectively. Our results are calculated from daily stepped 182-day windows (i.e. 6 months). With 365 trading days a year in the cryptocurrency market, we use δT = 1 day and T = 182 days. Each time window thus contains 182 daily returns. With these parameters, the overall number of windows is M = 2483. This procedure allows us to accurately

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reflect the structural changes of the cryptocurrency network using relatively small intervals between shifts, but at the same time, it can maintain sufficiently large sample sizes of the cryptocurrency pricing data for calculating the cross-return correlation coefficient matrix for each shift. Note that, as our starting date is 29.04.2013, there is only a few cryptocurrencies in our dataset. As time goes on, more cryptocurrencies have been progressively added into the market. In this case, in order to have a reasonable number of observations to draw meaningful conclusions, we consider the cross-return correlation coefficient between any two cryptocurrencies to be valid if and only if both of them have T = 182 daily return for a given time window; otherwise, the correlation would not be considered and thus no edge would be created in the network, e.g. in a given time window, cryptocurrency X and Y have respectively 80 and 182 daily returns since Y joins the market earlier than X. As such, there is no correlation coefficient and edge exist between X and Y in this time window.

V. Centrality measures

To analyzing a network, it is useful to use well-established centrality to identify important or influential cryptocurrencies that are located in the core part of the network. We use the following five commonly used centrality in this study [8].

A. Degree Centrality (DC)

The degree centrality of node v is simply the total number of its immediate connected neighbor nodes and is defined as:

$$c_D(v) = \frac{\sum_{u\epsilon V}^{N} A_{vu}}{N-1}, v \neq u \tag{4}$$

where V is the set of all nodes, A_{vu} is an adjacency matrix in which value is equal to 1 if v and u are connected to each other; otherwise it is equal to 0. The degree centrality values are normalized by dividing the possible maximum number of edges in a network N - 1 where N is the number of nodes. DC measures the relative significance or power of a cryptocurrency, in terms of the number of edges upon it, that can directly affect the other cryptocurrencies. A cryptocurrency node with a higher DC score is likely to influent the behavior of more other cryptocurrencies that are directly connected to it.

B. Node Strength (NS)

An extension to the degree centrality is to further consider the correlation in addition to the number of their immediate neighbor nodes since a larger number of immediate connected neighbor nodes may not indicate a stronger correlation with these neighbor nodes. The node strength of node v is the sum of the correlation of v (i.e. edges) with all other nodes to which it is connected and is defined as:

$$c_{ns}(v) = \sum_{u\epsilon V}^{N} c_{vu}\, A_{vu}, v \neq u \tag{5}$$

Node strength measures how strong a cryptocurrency is correlated with the others and integrates the information of both its connectivity and the significance of its edges in terms of their total correlation. This can be taken into account as the normal generalization of the connectivity.

C. Betweenness Centrality (BC)

The betweenness centrality of node v is the sum of the fraction of all-pairs shortest paths that pass through v:

$$c_B(v) = \sum_{s,t\epsilon V} \frac{\sigma(s,t|v)}{\sigma(s,t)} \tag{6}$$

where $\sigma(s,t)$ is the number of the shortest path between nodes s and t, and $\sigma(s,t|v)$ is the number of shortest paths passing through v between s and t. If $s = t$, $\sigma(s,t) = 1$, and if $v\epsilon(s,t), \sigma(s,t|v) = 0$. BC is used to quantify the control of a cryptocurrency node on information flow in the network. Therefore, cryptocurrency nodes with high scores are considered as significant cryptocurrencies in terms of their roles in coordinating the information among cryptocurrencies. Thus, the higher the value of $c_B(v)$, the more the node is likely to be an intermediate node or broker that transfers or coordinates information from one node to another through it.

D. Closeness Centrality (CC)

Closeness centrality of node v is the reciprocal of the sum of the shortest path distances from v to N - 1 other nodes. Since the sum of distances depends on the number of nodes in the network, closeness is normalized by the sum of minimum possible distances (i.e. $N - 1$) and is defined as:

$$c_C(v) = \frac{N-1}{\sum_{u\epsilon V}^{N} d_{vu}}, v \neq u \tag{7}$$

where d_{vu} is the shortest path distance between v and u. CC measures how close a cryptocurrency node to all other cryptocurrency nodes. The higher the score of a cryptocurrency node, the faster the cryptocurrency spread information from it to all others. Therefore, it represents the effectiveness of information propagation from a cryptocurrency to the others. This is based on an idea saying that if one cryptocurrency can reach others quickly, this cryptocurrency is at the central position in a network.

E. Eigenvector Centrality (EC)

Node degree may not be seen as a sufficient representation of how important or influential a node is, for it does not take into account how important the neighbors of a node may be. As an example, one node may have a low degree centrality, but it may be connected with other nodes with very high DC or NS, so it is, in some way, influential. A measure that takes into account the degree of neighboring nodes when calculating the importance of a node is called eigenvector centrality. In order to define it properly, one must first define an adjacency matrix, $\mathbf{A}$, whose element a_{ij} is 1 if there is a connection between nodes i and j, and 0 otherwise. The EC of node v is:

$$c_E(v) = \frac{1}{\lambda}\sum_{u\epsilon V}^{N} A_{vu} c_E(v), v \neq u \tag{8}$$

Compared to DC and NS, EC considers not only the number of or correlation strength to immediate neighbor nodes but also their importance. Thus, edges connecting to more important nodes will offer higher centrality in a node than edges connecting to less important nodes. EC measures the strength of a cryptocurrency node with its connections to each major subgroup of the cryptocurrencies. Hence, the cryptocurrency nodes with higher scores are connected to the high-scoring cryptocurrencies. Comparatively, the cryptocurrency nodes having equal connections to low-scoring cryptocurrencies contribute less to the scores of the cryptocurrencies. EC thus computes the centrality for a node based on the centrality of its neighbors.

VI. Evaluation

A. Basic Properties of the Cross-return Correlation Coefficient Distribution

The cross-return correlation coefficient c_{ij} reflects the correlation degree of the relative logarithmic return between

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cryptocurrencies *i* and *j*, while the distribution of the cross-return correlation coefficients shows the correlation distribution of the whole cryptocurrency market across the time period we consider. We conduct a time-varying analysis of the cross-return correlation coefficient distribution by its first four moments, namely the mean, the variance, the skewness and the kurtosis.

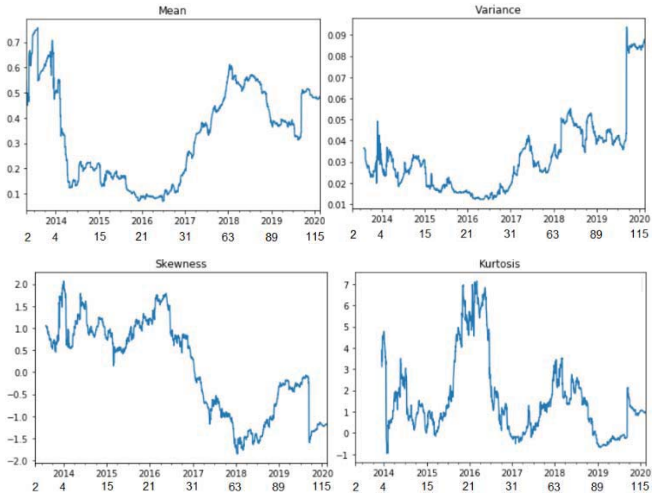

Fig. 2. The mean, variance, skewness and kurtosis of cross-return correlation coefficients of the 120 cryptpcurrencies in the market as a function of time

Figure 2 shows the first four moments of the cross-return correlation coefficient distribution over all the 2483 time windows. The numbers below the x-axis are the number of emerged cryptocurrencies from our dataset across different periods of time. In terms of the mean correlation, it exhibits a V-shaped structure in overall. Since the market in early 2013 is relatively "narrow", which has much fewer cryptocurrencies (e.g. BTC, Litecoin − LTC, Ripple − XRP and Dogecoin − DOGE in our dataset) than the number of cryptocurrencies exchanged today, they are generally correlated due to the strong influence of BTC on the others. More specifically, we observe a strong correlation between BTC and LTC (0.76 – 0.86) and a moderate correlation between BTC and XRP (0.35 – 0.5), and between BTC and DOGE (0.45 – 0.6), which keeps the overall mean correlation high in general, even though new cryptocurrencies other than those four mentioned above, which have low to moderate correlation with the others (0.25 to 0.5), are progressively added into the market. *This reveals that the early stage of the cryptocurrency market exhibited a medium to strong correlation among the cryptocurrencies.*

As time goes on, there is a sharp increase in the number of cryptocurrencies in the market since 2014 (66 in 2013 vs. 506 in 2014). There are approximately 6088 cryptocurrencies being traded with a total market capitalization of more than 338 billion USD (as of 05.08.2020) [17]. These cryptocurrencies are created and used for a wide variety of applications, e.g. some are being used for transaction payment, while some are created for blockchain applications in different industries such as healthcare and engineering. As shown by the valley in 2016 in Figure 2, the low mean correlation indicates weak relationship whatsoever between many pairs of cryptocurrencies, e.g. cryptocurrencies that are being used for finance and healthcare may not have any relationship with each other and thus their prices or return are likely to move individually. We observe that, during the valley period, the return correlation of about 80% of all the cryptocurrency pairs are very low, only between 0.0 and 0.2, except for a few pairs such as BTC and LTC. *This reveals that the middle stage of the cryptocurrency market exhibited a weak correlation among the cryptocurrencies, possibly due to the emergence of many new cryptocurrencies that are created for a wide range of applications without reciprocal effects from the business and technology perspectives such that they act independently and their returns move individually.*

Starting from mid-2016, more and more cryptocurrencies are increasingly correlated again. One of the reasons to explain this phenomenon is that, there is a crash in the cryptocurrency market as the price of BTC falls by about 65 percent during the month from 6 January to 6 February 2018 [9]. Subsequently, nearly all other cryptocurrencies also peaked from December 2017 through January 2018, and then followed BTC. By September 2018, cryptocurrencies collapses 80% from their peak in January 2018, making the 2018 cryptocurrency crash worse than the Dot-com bubble's 78% collapse (albeit with a vastly smaller amount of money invested). By 26 November, BTC also falls by over 80% from its peak, having lose almost one-third of its value in the previous week. This increase of the mean correlation value suggests that, during this crash period, all cryptocurrencies are so highly correlated that the entire cryptocurrency market is dragging down as a whole since any financial crisis often cause an increase of the market's cross-return correlations. In addition to this, the increasing correlation may indicate the increased synchronization among the cryptocurrencies' prices [5]. More specifically, many cryptocurrencies are becoming mature and are seen in the eyes of the investors as more homogeneous assets, becoming increasingly better substitutes of each other. Thus, their time series properties, i.e. return and volatility, are expected to converge with time and this is confirmed from the study in [10] that the increasing correlation is reflected in an exponentially decreasing number of isolated nodes and the creation of a denser network with growing integration and interconnectedness, in which more cryptocurrencies exhibit similar behavior.

Moreover, as the COVID-19 outbreak worsen since early 2020, the price of many cryptocurrencies dropped significantly in two days in mid-March, which further increased the mean correlation value. We conjecture that another reason for the increased correlation is that there are reciprocal effects among the underlying applications of the cryptocurrencies or the use of blockchain technology which is being widely adopted as a solution by various industries. In overall, compared to the past days, the correlation among cryptocurrencies today remains consistently strong. *Therefore, in the same way as the stock market, investors are advised to pay attention to the trends or news of other cryptocurrencies before making any trading decision on a particular one since a few of them remain being the influencers in the market and some are strongly correlated.*

In terms of the variance, it generally follows the movement of the mean correlation since the correlation between the mean and the variance is very high (~ 0.78). This implies that when the mean correlation increases, possibly due to some negative events in the market or society such as crash (in 2018) and COVID-19 epidemic (since early 2020), the variance increases. Thus, the dispersion of values of the correlation coefficient is higher. In terms of the skewness, it is inverse to the mean correlation as high (low) mean correlation implies

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that the distribution of the return correlation coefficients becomes more symmetric (asymmetric). In terms of the kurtosis, particularly the spike in 2016, a high degree of kurtosis (i.e. leptokurtic for which the kurtosis values are greater than the normal kurtosis 3) and a long right tail represents a fat-tail distribution that explains the phenomenon of having a very few cryptocurrency pairs with strong correlation (e.g. 0.8 between BTC and LTC), while majority of other pairs have relatively weak correlation (0.1 – 0.35).

TABLE I. CORRELATIONS AMONG THE FIRST FOUR MOMENTS USING PEARSONS 'S LINEAR AND SPEARMAN'S RANK-ORDER (VALUES IN THE PARENTHESIS) CORRELATION COEFFICIENTS

	Mean	Variance	Skewness	Kurtosis
Mean	- -	0.65 (0.78)	-0.70 (-0.70)	-0.34 (-0.24)
Variance	- -	- -	-0.69 (-0.73)	-0.33 (-0.38)
Skewness	- -	- -	- -	0.42 (0.29)
Kurtosis	- -	- -	- -	- -

Furthermore, we investigate whether these four moments are correlated by calculating their Pearson's linear and Spearman's rank-order correlation coefficients in Table I, the values of which between the mean value and the variance are 0.65 and 0.78, while the ones between the skewness and kurtosis are 0.42 and 0.29, respectively. Thus, the first two (i.e. mean and variance) and the last two (i.e. skewness and kurtosis) moments are correlated with each other, as can be observed from their co-movement in the figure. On the contrary, the mean-variance and the skewness-kurtosis are anti-correlated, which can be observed by their opposite co-movement in the figure and negative correlation coefficients in the table (i.e. when one moment increases, the other decreases, and vice versa).

B. Dynamic Structure of the Cryptocurrency Network

In this section, we investigate the stability of the cryptocurrency network constructed by MST across its evolution. We examine (i) the topological properties of the network, and (ii) the linkage survival in the network.

To analyze the topological properties of the cryptocurrency network, normalized tree length (NTL) is used to measure the average distance of the network at time t [13]:

$$NTL(t) = \frac{1}{N-1}\sum_{d_{i,j}^t \in E_t} d_{i,j}^t \tag{9}$$

where $d_{i,j}^t$ is the distance between cryptocurrency node i and j at time t, E_t denotes the set of all edges of the network at time t, and $N-1$ is the number of edges.

On the other hand, the average path length (APL) can be used to analyze the network density, which is defined as

$$APL(t) = \frac{2}{N(N-1)}\sum_{i<j} l_{i,j}^t \tag{10}$$

where $l_{i,j}^t$ is the shortest path between nodes i and j in the network at time t. In addition, the mean occupation layer (MOL) proposed by [14] is used to characterize the spread of nodes in the network, which also can be used to quantify the changes in the density of the network, with smaller values suggesting higher density and its nodes tend to crowd the area around the central node in the network. It is defined by

$$MOL(t, v_c^t) = \frac{1}{N}\sum_{i=1}^{N} Lev(v_i^t) \tag{11}$$

where v_c^t and v_i^t are the central node c and node i, respectively, in the network at time t, and $Lev(v_i^t)$ is the difference in level between nodes i and c at time t when the level of the central node c is set to zero. The central node can be either the node with the highest number of links or the highest sum of correlations of its links in the network. For each time window, we choose as the central node the one that minimizes the MOL. It is also noted that one can have either a static (fixed at all times) or a dynamic (continuously updated at each time window) central node. Since the cryptocurrency market is not dominated by only one cryptocurrency, we adopt the dynamic approach in this study. The MOL characterizes the compactness of the network, which usually shrinks during abrupt price changes (e.g. crisis).

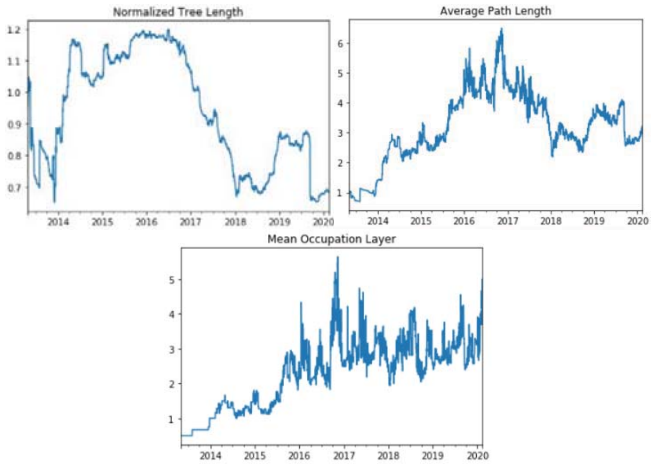

Fig. 3. Time-varying normalized tree length, average path length and mean occupation layer of the cryptocurrency network

Figure 3 shows the time-varying results of NTL in the network. The figure shows that, during the cryptocurrency crash in 2018 and the COVID-19 epidemic in early 2020, the NTL values are obviously at the lowest that forms two valleys. Specifically, low NTL corresponds to a strong correlation across the cryptocurrency market. *Similar to stock markets, the cryptocurrency market exhibits the behavior that crisis, either financial, epidemic or others, usually lead to an increase in correlations, which makes the cryptocurrencies more sensitive to changes in market price.*

Figure 3 also shows the results of APL and MOL. One can see large fluctuations since 2016 and it has the tendency to increase, indicating that the network over time is becoming less compact. We also find that the MOL reaches a lower value on a few occasions, which can be traced to the BTC stolen from Bitfinex in 2016 and crash in 2018 at which BTC prices fall dramatically. An economic interpretation is that during a market crisis, common factors should drive cryptocurrency returns instead of company- and industry-specific factors; therefore, a denser network indicates an increase in the importance of common factors [14]. The low MOL value during the crisis means that the transmission of price fluctuations and other information from the central node to other nodes requires fewer intermediate cryptocurrencies than during more stable periods. In contrast, high MOL values are considered to reflect a finer market structure, whereas in the other extreme low dips are connected to market crashes. This reveals that the network is becoming denser and more concentrated around the central node as well as the co-movement of stocks is reinforced as the market becomes

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volatile, which is consistent to our subsequent findings about the power-law distribution of the cryptocurrency network.

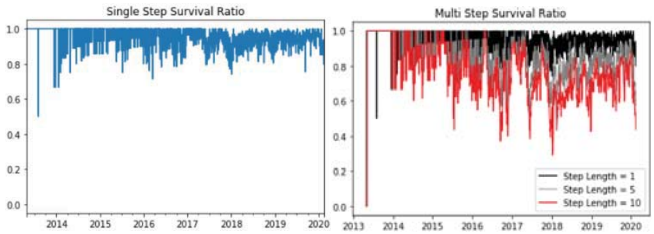

Fig. 4. Single-step and Multi-step survival ratio of the cryptocurrency network

We next use the single-step survival ratio (SSR) [14] to quantify the edge evolution of the cryptocurrency network:

$$SSR(t) = \frac{1}{N-1}|E(t) \cap E(t-1)| \tag{12}$$

where *E(t)* is the set of edges in the network at time *t*. The larger the SSR, the more stable the network in a short-time basis. Figure 4 shows the SSR of the network. The figure shows that the SSR trends are similar across the time and its values are very high (the average SSR is 0.95). This indicates that most linkage (on average 95% of the edges in the network) survive from one time to the next and that the network in the short term is stable.

To study a longer term evolution of the networks, we use the multi-step survival ratio (MSR) at time *t* as:

$$MSR(t,k) = \frac{1}{N-1}|E(t) \cap E(t-1) \cap \ldots E(t-k+1) \cap E(t-k)| \tag{13}$$

where *k* is the step length and only the edges that are constantly present in the network after *k* time steps survive. For small and large *k* values, *MSR(t, k)* quantifies the short-erm and long-term network stability, respectively. SSR(*t*) is simply a special case of *MSR(t, 1)*. We quantify the robustness of the edges by measuring the MSR using different step lengths. We set the step length to be 1, 5 and 10.

Figure 4 shows the results from 3 different MSTs. Note that the MST curves show the same trend across different step lengths. Note also that increasing the step length intensifies the MSR curve, suggesting that the edges in the network become less stable over time. For example, when the step length is 5 days, the average MSR is 0.85. If we increase the step length to 10 days, however, the average MSR drops to 0.76, suggesting that 76% linkages in the network is identical over 10 days. *This indicates that the cryptocurrency pairs with strong correlations are robust and remain stable across the studied periods. Thus, the trend movement of one cryptocurrency is likely to be influenced by or to influent its counterpart cryptocurrency that is highly correlated with it. Investors are advised to make timely adjustments in their portfolios if they are to minimize risk.*

C. Scale-Free Structure of the Cryptocurrency Networks

So far we have characterized the cryptocurrency network derived from all the elements of the correlation matrix. Since the MST is expected to reflect some aspects of the cryptocurrency market and its state, it is therefore of interest to learn more about its structure. During the last few years, much attention has been devoted to the degree distribution of networks, so called scale-free networks, where this distribution a power-law. Many real networks are found scale-free. If a node *i* has *k* edges in a network, then *k* is designated the degree of node *i*. The degree distribution *p(k)* is the probability that a node will have *k* edges and is also the ratio between the number of nodes with *k* edges and the total number nodes in a network If the degree distribution *p(k)* of a network has a power-law tail, i.e.,

$$p(k) \sim k^{-\alpha} \tag{14}$$

the network is scale-free or has a scale-free structure. To study the scale-free structure of the cryptocurrency networks, we use the analytical methodology proposed by Clauset et al. [18] that combines the maximum-likelihood estimation method of fitting a power-law with the Kolmogorov-Smirnov test for goodness-of-fit and confidence-interval.

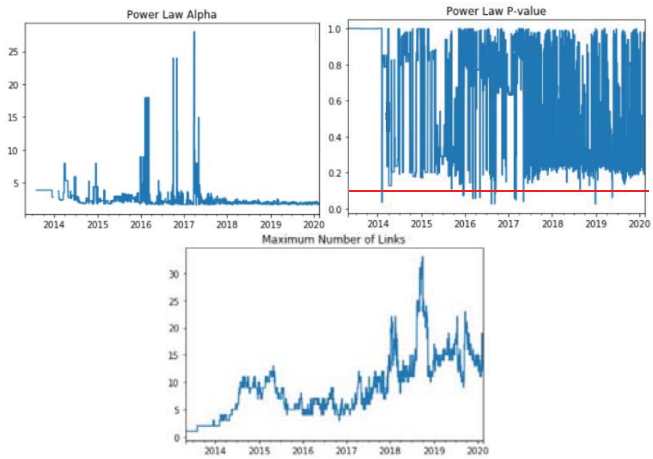

Fig. 5. Power law distribution of the cryptocurrency network

Figure 5 is a graph of the power-law distribution. Note that when the estimated power law p-value is greater than 0.1, the power-law hypothesis for the empirical data stands. Figure 5 shows that the average power-law exponent α is 1.66 and the corresponding p-values are mostly greater than 0.1. *This indicates that the cryptocurrency networks are scale-free networks that follow power-law* [18]. The closer the value of the power-law exponent α is to 1.0, the longer the tail of the distribution, and the larger the proportion of the distribution in the tail. Figure 5 shows that this power-law characteristic is also supported by the increasing maximum number of links of the most connected nodes in the networks.

D. Centrality Measures Evaluation

A node with large centrality may imply that it is important or influential in a network. The cryptocurrency nodes with the highest and the second highest centrality measures based on the MST network construction method are plotted in Figure 6. Cryptocurrency nodes that are not among the two highest centrality are not displayed in the figures to allow for better visualization of the influential cryptocurrencies. In general, these figures have reached similar results from which we can define 3 stages for the evolution of cryptocurrency market as highlighted by the three black dotted line boxes at the top of Figure 6, in which BTC dominates the first stage, followed by MaidSafeCoin (MAID) and Factom (FCT) in the second stage and finally ETH takes the leading position in the third or the most recent stage. Some characteristics of these 3 stages are summarized in Table II.

The first stage, from early 2013 to mid-2016, is mainly dominated by cryptocurrencies that are created for financial services. The most influential cryptocurrency is BTC, which

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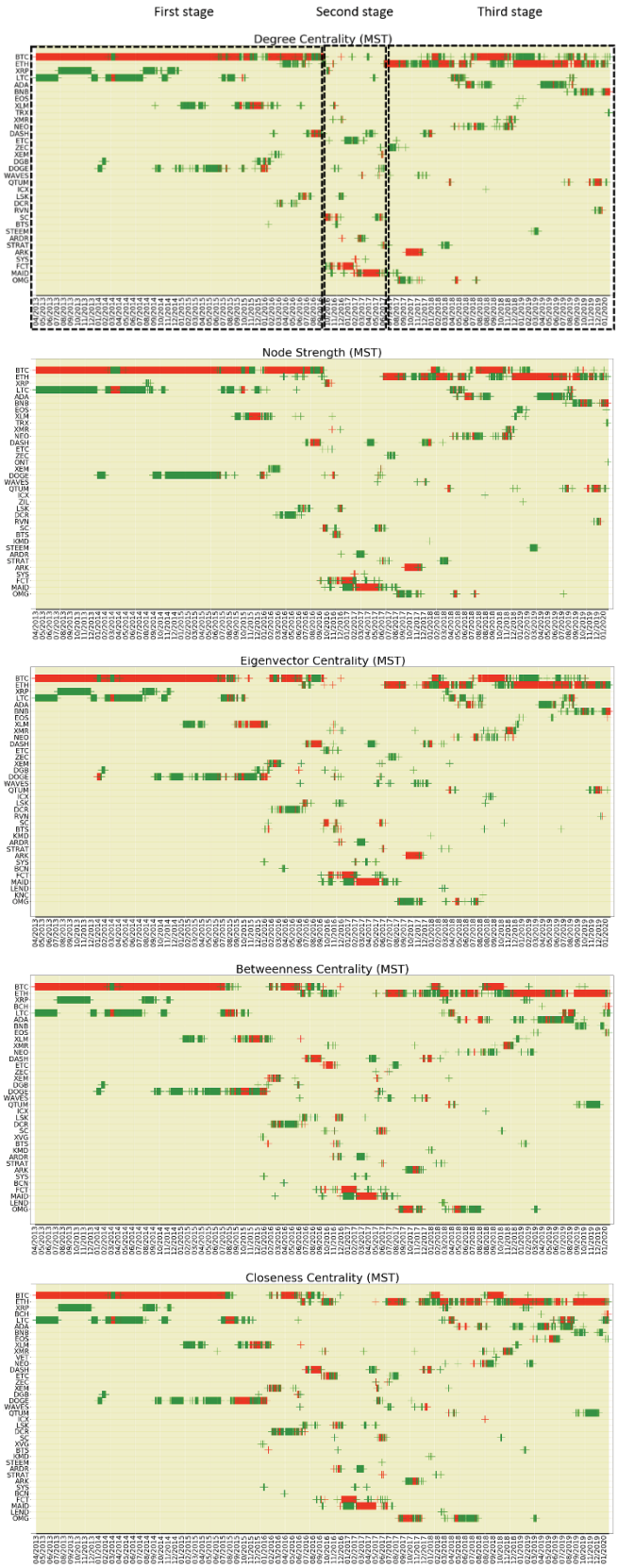

Fig. 6. Cryptocurrencies with the highest (red) and the second highest (green) centrality measures in the cryptocurrency MST networks

is primarily being used for transaction payment and mining. Other cryptocurrencies, such as LTC, XRP, XLM, DOGE and BTS, serve similar financial purposes as BTC does. This confirms the findings from [12] that there exists close relationship among BTC, LTC, DOGE, XRP, XLM and BTS

TABLE II. A SUMMARY OF 3 STAGES OF THE CRYPTOCURRENCY MARKET EVOLUTION

Stage	Cryptocurrencies with the two highest centrality measures	Main purpose of the cryptocurrencies
Early 2003 – Mid 2016	BTC, LTC, XRP, XLM, DOGE, BTS, DASH	Transaction payment
Mid 2016 – Mid 2017	MAID, FCT, SC, ARK	Cloud data storage and recording in blockchain networks
Since Mid 2017	ETH, ADA, NEO, OMG	Ethereum-based with smart contracts
During COVID-19	ETH, QTUM, BNB	Specific events

on their MST networks, with the former three being placed in the middle of the network. The work in [11] also confirms that BTC is the most influential cryptocurrency, followed by LTC which has significant influence on BTC as well as on other leading cryptocurrencies, which suggests that BTC starts losing its dominant role in the evolving cryptocurrency market (BTC's dominance in market capitalization has gradually faded from 85% in 2010 to 50% today [11]). The work in [10] finds that BTC and LTC are close to each other, and XLM, XRP, BTS and DOGE are within a community in the network.

The second stage, from mid-2016 to mid-2017, emphasizes the use of blockchain technology in developing applications, particularly cloud data storage and recording by SC, MAID and FCT which have been identified influential cryptocurrencies in this stage. This implies that cryptocurrencies with high market capitalization may not always be identified as influential (e.g. SC, MAID and FCT are ranked #71, #78 and #119, respectively in comparison to BTC which is ranked #1). MAID creates a secure through encryption and decentralized data-sharing networks, namely SAFE networks, using unused computer capacities such as disk space. MaidSafe users can become network nodes and receive some coins in rewards for contributing their unused disk space. Afterward, MaidSafe utilizes a vast amount of these unused disk space to offer a decentralized network storage service. Sia serves a similar purpose alike MAID. On the other hand, FCT creates an immutable and decentralized database working on electronic data notarization to enhance data integrity in data recording, e.g. financial, medical and voting data, in its blockchain. In 2016, Factom signed a smart city deal to roll out blockchain verification across China many cities and was working with other countries, e.g. Honduran, to create immutable land registry database. The work in [4] reveals that MAID and FCT are ranked amongst the top 7 by their net-connectedness and PageRank to other cryptocurrencies, while MAID being the center of the network during the period. The work in [10] find that MAID is one of the largest communities with respect to the number of cryptocurrencies in it, which also includes FCT and SC.

Finally, the third stage or the most recent stage since mid-2017 encourages adopting advanced blockchain features in applications, such as programmable smart contract which is a small computer program that "live" in the blockchain to allow various pre-determined rules of any activities to be encoded into unbreakable contracts and then automates the implementation and execution of these contracts when agreed upon conditions are met. This protects all parties involved, eliminates confusion, and has the potential to dramatically reduce verification, execution and legal costs. ETH is the platform that supports smart contracts in the blockchain environment. Furthermore, Blockchain-As-A-Service (BaaS) is a cloud-based service that allows customers to build their own blockchain-powered products including apps, smart contracts, and use other blockchain features without the need

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for setup, manage or execute blockchain-based infrastructure. Few companies including Amazon's AWS, Microsoft's Azure and others already provide this service. The use of BaaS is expected to increase in 2019 onwards. This further emphasizes the use of Ethereum as the foundation underlying technology for blockchain application development. Other cryptocurrencies, such as ADA, NEO and OMG, are closely related to ETH as can be observed by their strong correlation among each other (0.8 – 0.87).

Lastly, we consider closeness and betweenness centrality. The cryptocurrencies listed in Table II are the closest to and being passed through by other nodes in the network during the stages they belong to. Thus, cryptocurrencies among the highest rank in one centrality are likely to be the highest rank in most other centrality measures.

E. Cryptocurrency Centrality Measures Under COVID-19

In the previous section, we have identified some influential cryptocurrencies at different stages of the market, with ETH taking the leading position during the recent years. However, we find in Figure 6 (the rightmost part) that, since the outbreak of COVID-19 in early 2020, the most influential cryptocurrencies have been intermittently shifted to QTUM and BNB. First of all, QTUM, a platform that leverages ETH smart contract functionality along with the security of BTC network, has recently published its Q2 2020 report in which their most requested feature of Offline Staking code is released to the community, which allows users to stake their QTUM from mobile, hardware and web wallets [15]. With the new objective of moving a lot of focus to the community, QTUM has also increased its online or social media community engagement, due to COVID-19 and many people staying home because of lockdowns, by hosting many Ask Me Anything sessions, quizzes and key development updates. On the other hand, BNB has recently launched a $5 millions COVID-19 relief champaign "Crypto Against COVID" by donating financial and medical resources to many countries in the world [16]. We believe that these COVID-19 related or aided events would have influential impact in the community, which makes QTUM and BNB important cryptocurrencies that have attracted increasing public attention in recent months (their price movement have exhibited uptrend). As a result, *investors are advised to pay attention to those cryptocurrencies that provide relief aids to the COVID-19 pandemic before making any trading decisions as the cryptocurrencies may benefit from the epidemic outbreak.*

VII. Conclusion

This paper studies the evolution of the cryptocurrency market using network analysis and centrality measures on the time series daily close price of top 120 cryptocurrencies between 2013 and 2020. Our study has two primary findings: (1) the overall cross-return correlation among the cryptocurrencies is weakening from 2013 to 2016 and then strengthening thereafter; (2) cryptocurrencies that are primarily used for transaction payment, notably BTC, dominate the market until mid-2016, followed by those developed for applications using blockchain as the underlying technology, particularly data storage and recording such as MAID and FCT, between mid-2016 and mid-2017. Since then, ETH, alongside with its strongly correlated cryptocurrencies, such as ADA, NEO and OMG, have replaced BTC to become the benchmark cryptocurrencies, primarily due to its prevailing smart contract capability that can automate the execution of business activities written in the contracts. Furthermore, during the outbreak of COVID-19, QTUM and BNB have intermittently replaced ETH to take the leading positions due to their active community engagement during the pandemic. As future work, we apply network analysis and centrality measures to understand the evolution and characteristics of existing cryptocurrency tokens.

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